DAY 26 | PRE-READ

# Tools, Flows and Loops

A light read before Day 26. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

## Why read this?

So far you have asked a model one question at a time. Real assistants need several steps: look something up, calculate, check, and sometimes ask a person. Today you learn how to connect those steps safely.

## 1. Recipes for AI

A recipe lists steps in order: chop, fry, season, serve. A **chain** does the same with AI: fill a prompt, call the model, tidy the answer. Frameworks such as LangChain give you ready-made steps that snap together.

## 2. Letting the model use tools

A support agent on the phone looks up your order in a system before answering. A model can do the same. We describe our functions as **tools**; the model asks for one, and our code decides whether to run it. The model never presses the button itself.

## 3. Every tool costs words NEW (v4)

Each tool's description is sent to the model on every call, and each tool result is read by the model. Many long tool descriptions mean many **tokens**, so more cost. Good tools are few, clear and return only what is needed.

## 4. Flowcharts that remember

Some work is a straight line; some has branches and loops: "if the answer is weak, search again". A **graph** describes those branches. **LangGraph** runs such graphs and keeps a shared notebook of facts called **state**.

## 5. Pause, ask, continue NEW (v4)

Before refunding money, a careful assistant stops and asks a manager. A graph can **pause** at that step, save everything, and continue hours later with the manager's answer. Saving progress like this is called **durable execution**.

| Everyday idea | Technical term |
|---|---|
| Recipe | Chain |
| Phone lookup | Tool call |
| Flowchart | Graph |
| Shared notebook | State |

| Everyday idea | Technical term |
|---|---|
| Save game | Checkpoint |
| Ask the manager | Human-in-the-loop interrupt |

**Key idea:** the model suggests; your code decides. Risky actions always need a human yes.

## 6. Think about these before class

- Which actions in a telecom support team should never happen without a person approving them?
- If an assistant had 30 tools, how could it pick the right one quickly and cheaply?
- What should happen if the system crashes halfway through a refund?

Nagare (流れ): flow